

## Signals recap ~~for~~ Lecture 2

Why do we need complex valued signal? When signal are real? are transported

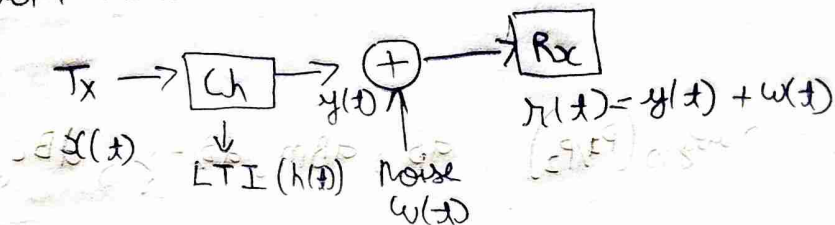
~~because analysis~~ because analysis becomes easier.

$x(t) \rightarrow x(\omega)$  complex has a complex representation

Sampling of continuous signals is lossless if signal is bandlimited.

After Sampling, you do quantization (Acts as a channel) which introduces random noise

↳ additive noise



noise is gaussian due to central limit theorem.

We assume channel is LTI, so we can characterize it using  $h(t)$ . Why is this assumption valid? (It is an approximation)

If  $y(t) = x(t) * h(t)$  what is the bandwidth of  $y(t)$

$$y(t) \leq \min(B_1, B_2)$$



We assume we have Energy signals and not Power signals in this course Page 61 of textbook

$f$  = frequency,  $|x(f)|^2$  energy spectral density  
 in a band  $\int_{f_1}^{f_2} |x(f)|^2$  ↳ used to reduce noise in or something on oscilloscope  
 axis ↳ check again

TDMA (Time Division Multiplexing), TDD

FDMA (freq division Multiplexing)

SDMA (Space) → antenna directivity → cellular systems

We want to reduce energy spectral density in other bandwidths  
 ↳ for example in FDM it might interfere with other bandwidth channels where other people may be communicating

In all of these Multiplexing we want to ensure the signals transmitted are orthogonal.

If something is orthogonal in time is it also in freq → check

$$\int x_1(t) x_2^*(t) dt = 0$$

CDMA  $\rightarrow$  Code division Multiplexing  $\rightarrow$  phase difference ( $\sin, \cos$ )

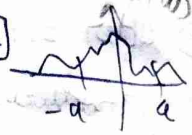
### Band limited Signals

we can never get Band limited signals  $\rightarrow$  means  $\infty$  time domain  $x(t)$

we can only generate time limited signal  $\rightarrow \infty$  frequency spectrum component

Definitions of bandwidths  $\rightarrow$

90% band width  $\rightarrow$  90% of energy spectral density  $[-a, a]$



Receivable:  $10 \log_{10}(P_1/P_2)$ , dB, dBm, dB/Hz, dBc

$\downarrow$   
mW

$\rightarrow$  carrier freq

3dB bandwidth  $\rightarrow$  where our power becomes half

Raghu Paley - Wiener Criterion,  $\rightarrow$  to check if a filter is realizable

$$\int_{-\infty}^{\infty} \frac{\ln |H(\omega)|}{1+\omega^2} d\omega < \infty$$

### Lecture 3

Time varying convolution  $\rightarrow$  for LTV systems.

We model communication systems as  $r(t) = h(t) * m(t) + n(t)$

$\downarrow$   
Receiver

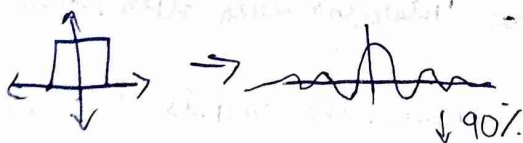
$\downarrow$   
Channel

$\downarrow$   
AWGN

message

We use orthogonality to separate carrier signals, so by taking their projection  $\int (m_1(t) + m_2(t)) m_2^*(t) dt$

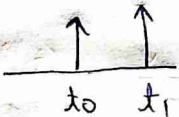
In real life we can only generate Linear phase systems filters



1) All types of filter that we use we put them into the channel.  
 ↓  
 at the receiver or transmitter

2) A simple wired channel can be modeled as  
 $h(t) = a_0 \delta(t - t_0)$  → attenuation and delay due to distance

3) But sometime due to reflections from objects in a wired communication we get -  $h(t) = a_0 \delta(t - t_0) + a_1 \delta(t - t_1) \dots$

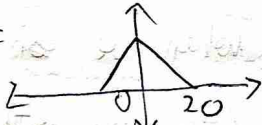
we get let say  → in time we get maxima and minima.

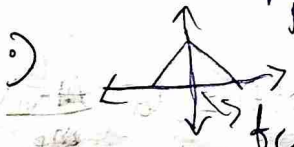
4) This is called Inter Symbol Interference (ISI)

5) We can construct Ideal filters in digital domain → why? because we made an assumption that  $x[n]$  is periodic (but we look at one period) so we see ~~the~~ sharp spectrum.  
 (Check)

Lecture 4

Recap  
 → we use GHz freq because of antenna length constraints.

6)  $M(\omega) =$   → we want to convert this to a higher freq for transmission → up conversion

7)   $= M(f - f_c)$  (carrier frequency)

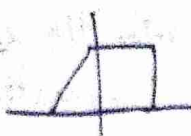
8) The equivalent representation is  $m(t)e^{j2\pi f_c t}$

9)  $m(t)$  can be complex as well,  $m(t) = m_{re}(t) + j m_{im}(t)$

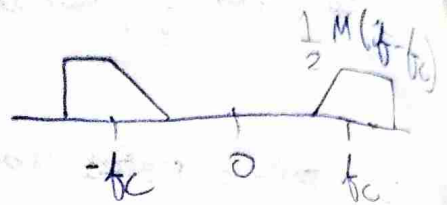
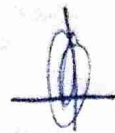
advantage, for a real signal the spectrum is symmetric so the info on one side is redundant.

 redundant so you

Can combine signal,  $m_1 =$    $m_2 =$    
 $\therefore m_1 + j m_2 =$  

1)  $M(\omega) =$  

Scale  $\left(\frac{1}{2}\right) M(-f - f_c)$

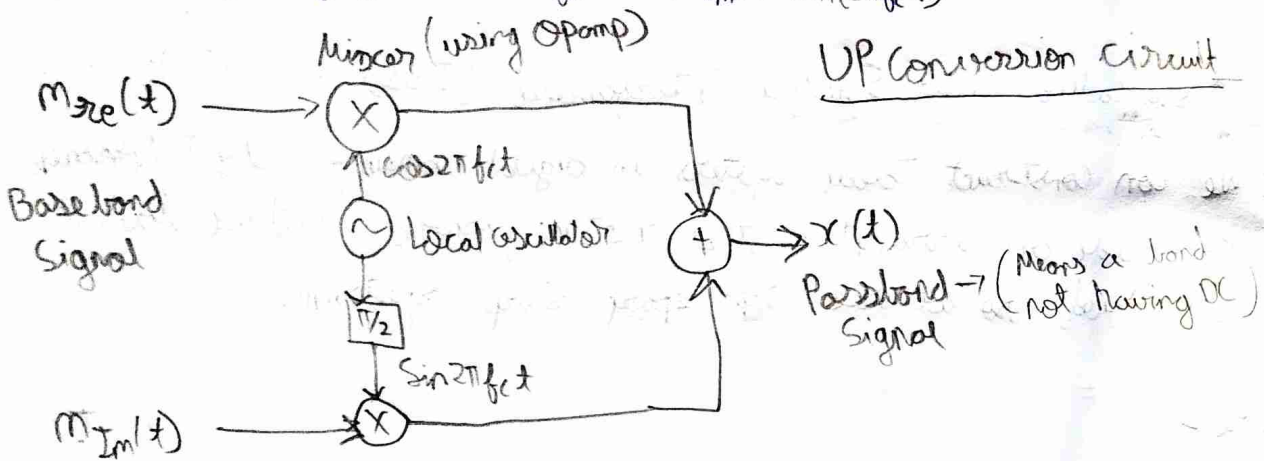


$x(t)$

for  $\text{Re} \{ m(t) e^{j2\pi f_c t} \} = \frac{z(t) + z^*(t)}{2}$ ,  $z(t) = m(t) e^{j2\pi f_c t}$

$M^*(t) \rightarrow m_{re}(t) \cos 2\pi f_c t + m_{im}(t) \sin 2\pi f_c t$

$x(t) = m_{re}(t) \cos 2\pi f_c t - m_{im}(t) \sin(2\pi f_c t)$



$\cos 2\pi f_c t \perp \sin(2\pi f_c t)$  (orthogonal), when we are modulating amplitude of both of them (they are orthogonal)

The phase is also being modulated

We define an envelope  $e(t) = \sqrt{m_{re}^2(t) + m_{im}^2(t)}$ ,  $\phi(t) = \tan^{-1} \left( \frac{m_{im}(t)}{m_{re}(t)} \right)$

$\therefore m_{re} = e(t) \cos \phi(t)$ ,  $m_{im} = e(t) \sin \phi(t)$

Substitute in  $x(t)$

to get  $x(t) = e(t) \cos(2\pi f_c t + \phi(t))$

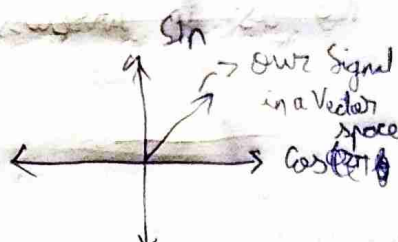
1)  $m_{re}$  and  $m_{im}$  can be arbitrary in time domain

2)  $\omega_m \text{ freq} > f_c \gg B$  because high bandwidth will spread to a freq which our antenna will not transmit, so we have this restriction.

$$m(t) = m_I(t) + j m_Q(t)$$

Inphase

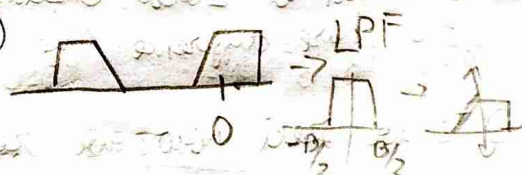
Quadrature Phase



3) Now we have to ~~convert~~ invert this signal

$X(f)$

$X(f + f_c)$



4) But  $\otimes$  IRL we cannot multiply with imaginary freq

we do  $\text{Re}(2x(t)e^{j2\pi f_c t})$

Scaling

$$x(t) \Rightarrow 2x(t)\cos 2\pi f_c t = (1 + \cos 4\pi f_c t) m_I(t) = m_I(t) + m_I(t)\cos 4\pi f_c t$$

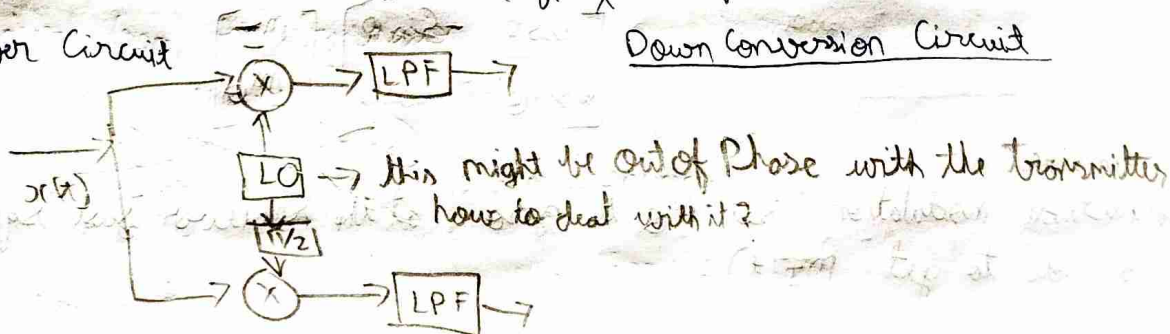
LPF  $\rightarrow m_I(t)$

to get other component,  $-2x(t)\sin 2\pi f_c t$

$$-2x(t)\sin 2\pi f_c t = -2m_I(t)\sin 2\pi f_c t \cos 2\pi f_c t$$

Receiver Circuit

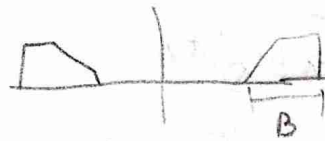
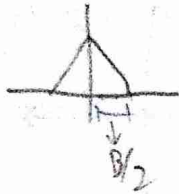
Down Conversion Circuit



Because we can have finite duration signals, our spectrum will be  $\infty$ , so we ensure 99% of energy in the region of interest. But if we have multiple antennas talking then 1% from each may add up.

## Lecture 5

- 1) If we use wired medium we are sending only 1 signal, are we sending less information ~~compared~~ compared to wireless? No as in wireless we use more bandwidth to send 2 signals



- 2) Both the upconversion and downconversion blocks are exactly the same but how are we able to recover the signal.  $\rightarrow$ ?

$\hookrightarrow$  Think about this

- 3) We assumed Local Oscillators are synchronized, if not then we get a phase difference (we lose orthogonality)

- 4) ~~If we~~ Now we receive  $x(t)\cos(2\pi f_c t + \theta)$

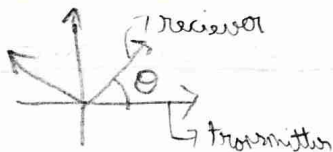
- 5) Transmitter power amplifier: multiply  $x(t)$  at the transmitter by 2 to avoid fading at the receiver (noise is amplified)

$$m_I(t) [\cos(4\pi f_c t + \theta) + \cos\theta] \\ - m_Q(t) [\sin(4\pi f_c t + \theta) + \sin\theta]$$

after ~~LPF~~ LPF, I  $\rightarrow m_I(t)\cos\theta - m_Q(t)\sin\theta$

$$Q \rightarrow m_I(t)\sin\theta + m_Q(t)\cos\theta$$

~~At~~ we receive a rotated axis at the receiver



$$\begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} m_I \\ m_Q \end{bmatrix}$$

Amplitude modulation, set  $m_Q$  to zero, at the receiver just square and add to get  $m_I(t)$

- 6) Phase modulation, set  $e(t)$  to 1 and only change  $\phi(t)$ , derivative of  $\phi(t)$  is the freq (freq modulation)

$\hookrightarrow$  we can use ~~phase~~ phase locked loops to synchronize

## Heterodyne = Super or Superheterodyne

- 1) AM our phones have too many bands of ~~freq~~ freq that they use. We get the carrier freq to an intermediate freq which is what our circuits are designed for.



$$y(t) = \text{Re} \{ x(t) e^{j2\pi(f_c - f_{IF})t} \}$$

do

- 2) There was an issue with circuit that freq at  $f_c - 2f_{IF}$  was also shifted to the  $f_{IF}$  freq, ~~then~~ you put a BPF In old AM radio stations they could hear 2 stations simultaneously.
- 3) At the end you put a BPF at  $f_{IF}$

## Lecture 6

- 1) If LO are synchronized then you call it homodyne
- 2) Superheterodyne  $\rightarrow$  Bringing down to intermediate frequency and then bring that down again to ~~the~~ DC.
- 3) ~~The~~  $f_{IF}$  is around 450 MHz

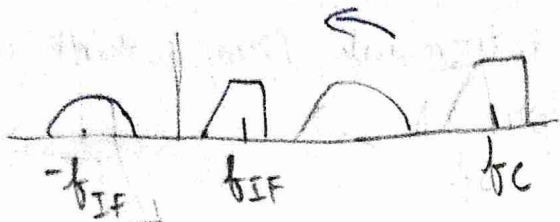
at the receiver side we have  $x(t) \cos(2\pi f_{LO} t)$

$$m_{\pm}(t) = \frac{m(t)}{2} [\cos(2\pi(f_{LO} + f_c)t) + \cos(2\pi(f_c - f_{LO})t)]$$

$$m_0(t) = \frac{m(t)}{2} [\cos(2\pi(f_c + f_{LO})t) + \cos(2\pi(f_c - f_{LO})t)]$$

we get  $f_c - f_{LO} = f_{IF} \rightarrow$  we convert to  $f_{IF}$

• We saw that this freq of  $f_{LO}$  we had issues



this is for  $\text{Re} \{ x(t) e^{j2\pi f_{LO} t} \}$

$$f_2 - f_{LO} = -f_{IF}$$

$$f_2 = f_{LO} - f_{IF} = f_c - 2f_{IF}$$

Image frequency.

• here we are shifting to the ~~right~~ left, but we can also shift to the right.

$$\hookrightarrow f_c + f_{LO} = f_{IF} \Rightarrow f_{LO} = f_c + f_{IF}$$

we want we get

• Why is it getting flipped?  $\rightarrow$  see complex representation

Spectral inversion

$$f_{LO} = f_c - f_{IF}$$

$$\begin{aligned} & \frac{m(t)}{2} \cos \\ & - m_Q(t) \sin \\ & = \frac{1}{2} \text{Re} \{ m(t) e^{j2\pi f_c t} \} \end{aligned}$$

$$f_c + f_{IF}$$

$$\frac{m(t)}{2} \cos$$

$$+ \frac{m_Q(t)}{2} \sin$$

$$= \frac{1}{2} \text{Re} \{ m^*(t) e^{j2\pi f_c t} \}$$

• How to select from the above two

we had image frequency as  $f_c - 2f_{IF}$  before and now shifting right

$$\text{we have } f_c + 2f_{IF}$$

$\hookrightarrow$  so we can use a low pass filter

it is an Image Rejection Filter.

• let's say  $f_c \in [f_L, f_U] \rightarrow$  some antenna, different  $f_c$  operation

picking  $LO, 1$

$$f_{LO} = [f_L - f_{IF}, f_U - f_{IF}]$$

$$= [f_L + f_{IF}, f_U + f_{IF}]$$

$$\rightarrow \frac{f_L - f_{IF}}{f_U - f_{IF}}$$

the ratio will be different

- That ratio is the swing of the filter and we want it to be small, even if the range of freq is the same.

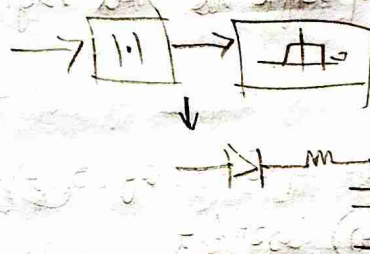
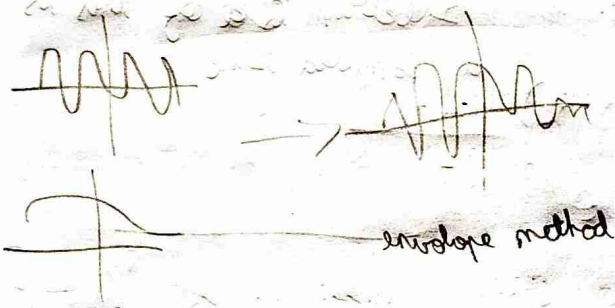
## Amplitude Modulation

•  $x(t) = e(t) \cos(2\pi f_c t + \phi(t))$

- Let's say we don't want to worry about  $\phi$  (Phase offset between filter). We can just set  $m_\phi(t) = 0$

then  $x(t) = m_I(t) \cos(2\pi f_c t)$

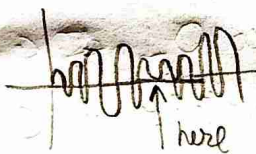
- AM is very inefficient, what we transmit are sending 4 times of what we need



- Issue with AM  $\Rightarrow m_I(t)$  cannot be -ve as 1.1 ignores it.

## Lecture 7

Phase reversal issue, for -ve  $m(t)$  there is a ~~phase~~ signal gets a discontinuity



DSB-SC

Double side band & Suppressed carrier

- we can add an offset to  $m(t)$

$\hookrightarrow (m(t) + A_c) \cos 2\pi f_c$ , with  $m(t) + A_c \geq 0$  always

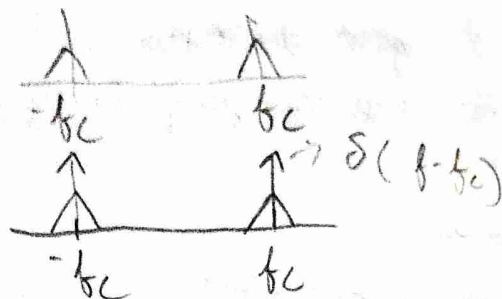
Amplitude of the carrier freq (DSB-AM)  $\nearrow$  Amplitude Modulation

- If  $\min m(t) = -M$ , then modulation index  $\alpha = \frac{M}{A_c} \leq 1$

$\hookrightarrow$  Also assume  $m(t)$  has no DC components

$\therefore A_c (1 + \alpha m(t)) \cos(2\pi f_c t)$

1) DSB-SC spectrum



2) PSB-AM spectrum

3) the modulus is not an LTI system, but how do you see it mathematically

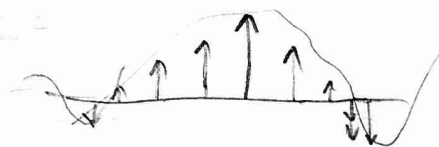
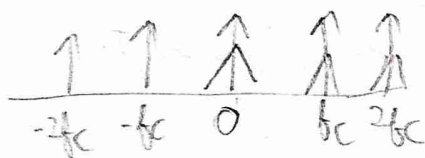
$x(t) \rightarrow$  [sine wave] ,  $|x(t)| =$  [rectified sine wave] equivalent to multiplying with [square wave]

which is specific to this signal only.

Spectrum of this is a sampled sine

so what is the spectrum of  $x(t)$

$$A_c(1 + \alpha m(t)) \cos 2\pi f_c t$$



4) How inefficient is our AM? How to quantify it?

$\hookrightarrow$  Power of message

Power of Tx signal

$$A_c \propto m(t) \quad A_c$$

$$\begin{aligned} x^2(t) &= A_c^2 (1 + \alpha m(t))^2 \cos^2(2\pi f_c t) \rightarrow \text{message} + \text{carrier} \\ &= \underbrace{\frac{A_c^2}{2} (1 + \alpha m(t))^2}_{y(t)} (1 + \cos(4\pi f_c t)) \end{aligned}$$

$y(t) \cos 4\pi f_c t$  after integrating is zero.

5) Also our assumption that the bandwidth of message is less than  $f_c$  so we will definitely not get a DC component  $\rightarrow$  Integral is zero

1) The power of  $x(t)$  is then  $\frac{A_c^2}{2} (1 + \alpha^2 P_m)$   
 Power of  $m(t)$  zero why?

2) Is <sup>Power</sup> ~~energy~~ in Passband the same as in the power in the baseband?  
 ↳ ~~part is~~ but it doesn't make AM or SC any efficient

$$x_m^2(t) = (A_c \alpha m(t) \cos 2\pi f_c t)^2$$

↓  
 this is in the Passband =  $\frac{A_c^2}{2} \alpha^2 m^2(t)$   
 has to be normalized

$$\eta = \frac{\text{Power of msg}}{\text{Power of Tx signal}} = \frac{\frac{A_c^2}{2} \alpha^2 P_m}{\frac{A_c^2}{2} (1 + \alpha^2 P_m)} \approx \frac{\alpha}{1 + \alpha} \rightarrow \text{can never be } 100\%$$

↓  
 Power efficiency

$$\text{PAPR} = \frac{\text{Peak of } m(t)}{\text{Avg of } m(t)} \rightarrow \text{it is energy and not Power (Misnomer)}$$

↓  
 Peak to average Power Ratio

1) maximum efficiency is (50%) but for this we assume  $P_m = 1$  and  $\alpha = 1$ , but if you send a sinusoid  $P_m = 1/2$  and then you get even lower efficiency. This just shows there is a relation to PAPR

## Lecture 8

1) we had assumed,  $\int m(t) = 0$ , no DC component.

2) How does the PAPR affect the whole of above setup.

$$\text{PARP} = \frac{\frac{A_c^2}{2} (1+1)}{\frac{A_c^2}{2} (1 + \alpha^2 P_m)}$$

High PAPR is always bad

$$\rightarrow \frac{2}{1 + \alpha^2 P_m} = 1.6$$

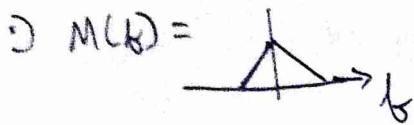
Power in  $A_c (\alpha m(t) + 1) \cos 2\pi f_c t$

now sub this in  $\eta = 1 - \frac{1}{2}$ ,  $\uparrow \text{PAPR}, \downarrow \eta$

SSB (Single side band)



removing redundancy.



$$M(f) = M(f)u(f) + M(f)u(-f)$$

$$M_+(f) + M_-(f)$$

$$m(t) = m_+(t) + m_-(t) \quad \text{because fourier transform is linear}$$

multiplying by  $u(f)$  is convolution in time  
↳ find out

$$X(f) = M(f-f_c)u(f-f_c) + M(f+f_c)u(f+f_c)$$

$$\therefore x(t) = m_+(t)e^{j2\pi f_c t} + m_-(t)e^{-j2\pi f_c t}$$

$x(t)$  is real signal  $\rightarrow$  verify,  $x(t) = x^*(t)$

$$x(t) = m_+(t) [\cos 2\pi f_c t + j \sin 2\pi f_c t]$$

$$m_-(t) [\cos 2\pi f_c t - j \sin 2\pi f_c t] \quad m_n(t)$$

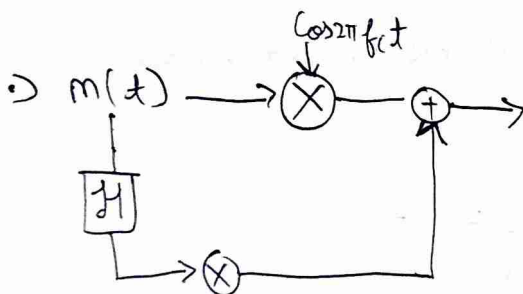
$$= m(t) \cos 2\pi f_c t - (-j m_+(t) + j m_-(t)) \sin 2\pi f_c t$$

In phase component

Quadrature

$$m_n(t) = \mathcal{H}(m(t)) \rightarrow \text{frequency response } -j \text{ sign}(f)$$

In frequency  $-j M_+(f) + j M_-(f) = 0$



find inverse transform of  $-j \text{ sign}(f) \leftrightarrow \frac{1}{\pi t}$

2) We can ~~also~~ send out message compl

3) Now analyse  $(m(t) + A_c) \cos 2\pi f_c t - m_n(t) \sin 2\pi f_c t$

$$= e(t) [\cos(2\pi f_c t + \phi(t))]$$

$$e(t) = \sqrt{(m(t) + A_c)^2 + m_n^2(t)} \approx A_c + m(t) \rightarrow \text{hilbert component being negligible}$$

4)  $m(t)$  is bandlimited so  $m_n^2(t)$  is not unbounded

makes us use envelope method seen previously to demodulate the signal

InSSB

5) The demodulation circuit is the same as one we saw for Down conversion, we get  $m(t)$  in the inphase arm, we are just sending a different looking signal.

QAM  $\rightarrow$  Quadrature Amplitude modulation

$\hookrightarrow$  2 AM signals which are orthogonal to, this is widely used

$\hookrightarrow$  Amplitude Shift Keying

Assignment 1



$\hookrightarrow$  find out why the spectrum was coming up with imaginary?  
 $\hookrightarrow$  indexing of fft was wrong - you had to shift the time domain signal too

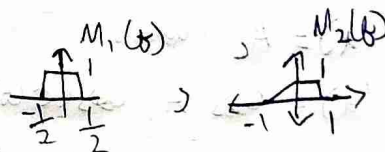
6) 3dB bandwidth  $\rightarrow$  log plot of energy spectral density, look at 3dB below the peak amplitude



Assignment 2

(I) Upconversion

Generate baseband signal by taking iFFT of



Upconvert to  $f_c = 10\text{Hz}$

plot the converted time and frequency

2) Downconversion

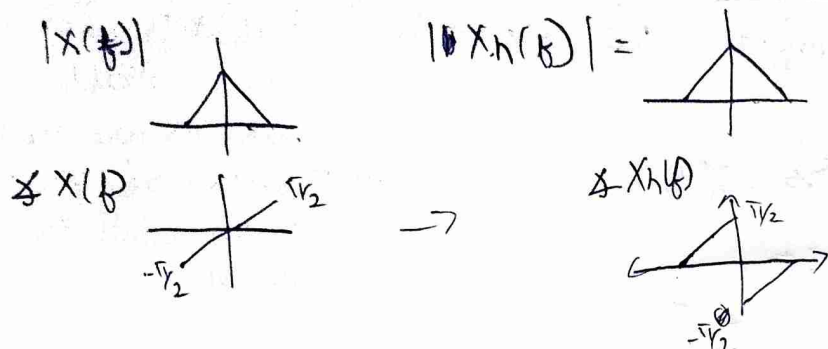
Downconvert your upconverted signal.

QAM  $\rightarrow x(t) = A_c [m_I(t) \cos 2\pi f_c t - m_Q(t) \sin 2\pi f_c t]$

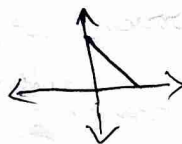
Hilbert transform

$H(x(t)) = x(t) * \frac{1}{\pi t}$

$X_h(f) = X(f)(-j \text{Sign}(f))$



Spectrum of  $x(t) + jx_h(t) \rightarrow$  analytic representation of a signal  
 $= X(f) + X(f)\text{Sign}(f) \rightarrow$



this is a technique to remove -ve freq values in spectrum

After this you upconvert it,

Section 2-8.3 about complex baseband is interesting  
 ↳ in text book

## Angle Modulation

We will work on  $\phi(t)$ , and we have 2 options

Phase Modulation (PM)  $\phi(t) = K_P m(t)$   $\rightarrow$  deviation constant

freq Modulation (FM)  $\rightarrow f(t) = \frac{1}{2\pi} \frac{d\phi}{dt} = K_f m(t)$

$\phi(t) = 2\pi K_f \int_0^t m(\tau) d\tau$

if  $c(t) = A_c$  (constant)

$m_I(t) = A_c \cos(\phi(t))$

$m_Q(t) = A_c \sin(\phi(t))$

If LO is not synchronized we get

$\cos\theta \cos\phi(t) + \sin\theta \sin\phi(t)$  in the I arm  
~~cos~~ ~~sin~~

we receive  $\cos(\phi(t) - \theta)$  and  $\sin(\phi(t) - \theta)$  <sup>Phase delay</sup>

we can do  $\cos^{-1}$  etc to recover phase  $\rightarrow$  which contains our message

$\Rightarrow$  this is good for PM, but how do you recover for F.M

Assumption  $m(t)$  has no DC component

$\Rightarrow$  you differentiate  $\frac{d}{2\pi dt}(\phi(t) - \theta) \Rightarrow \frac{d}{2\pi dt}(\phi(t)) \rightarrow$   $\theta$  just goes away.

so we prefer FM

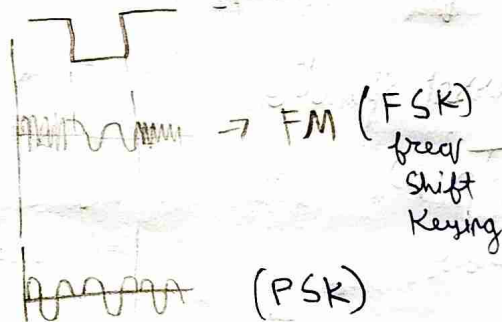
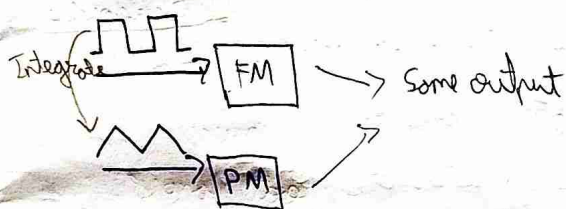
$\Rightarrow$  Circuits for Both FM and PM modulations work for each other

~~PM~~  $m(t) \rightarrow \boxed{\text{FM}} \rightarrow x_{\text{FM}}(t)$

$m(t) \rightarrow \boxed{\frac{d}{dt}} \rightarrow \boxed{\text{FM}} \rightarrow x_{\text{PM}}(t)$   $K_p = 2\pi K_f$

$m(t) \rightarrow \boxed{\int dt} \rightarrow \boxed{\text{PM}} \rightarrow x_{\text{FM}}$

$\Rightarrow$  What happens to the spectrum after modulation?



$\Rightarrow$  Satellite ~~using~~ use FSK, digital communication ~~uses~~ uses PSK

$\Rightarrow$  Modulation index  $\beta = \frac{\text{max}(m(t))}{\text{Bandwidth of the message}} \times \frac{d}{dt} \text{max}(m(t)) = \frac{\Delta f}{B}$   $\rightarrow$  max freq deviation  
 $= K_f \text{max}(m(t))$   $\rightarrow$  For PM  
 $B \rightarrow$  Bandwidth  
 $\rightarrow$  Max deviation of freq from the carrier freq

1) looking at spectrum of **FM**, PM is the same

2) Narrow band FM:  $\rightarrow \Delta f$  is very small,  $\phi(t) \ll 1$

$$A_c \left[ \underbrace{\cos \phi(t)}_{\approx 1} \cos 2\pi f_c t - \underbrace{\sin \phi(t)}_{\approx \phi(t)} \sin 2\pi f_c t \right]$$



this is again similar to A.M

$X_{FM}(f)$  has bandwidth as  $2B$ , we assume bandwidth is same even after integration

3) we have to use spectrum of  $\text{Re} \left\{ e^{j2\pi f_c t + j2\pi k_f \int_0^t m(t) dt} \right\}$

Say  $m(t) = \cos(2\pi f_m t)$  Bandwidth  $B = f_m$

we have  $\beta = \frac{k_f \times 1}{B}$  (modulation index)

$$e^{j\beta \sin 2\pi f_m t}$$

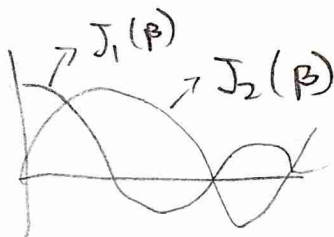
4) This signal is periodic

$$C_n = \frac{1}{2\pi} \int_{-\pi/2}^{\pi/2} e^{j(\beta \sin \omega - n\omega)t} dt$$

$$2\pi f_m = \omega$$

Bessel function

$J_n(\beta)$   
↓  
order



5) What we are transmitting  $\text{Re} \left\{ e^{j2\pi f_c t} \sum J_n(\beta) e^{jn2\pi f_m t} \right\}$

$$= \sum J_n(\beta) \cos(2\pi f_c t + n2\pi f_m t)$$

Assignment-2, Q7, w1

6) Message value goes to  $\infty$

7) do we want  $\beta \ll 1$

8) Do we want less bandwidth for FM because compression

• New class - Fourier series of  $\phi(t)$

$$c_n = \frac{1}{T_m} \int_0^{T_m} e^{j\beta \sin 2\pi f_m t} e^{-j2\pi n f_m t} dt \quad V = 2\pi f_m$$

$$\Rightarrow J_n(\beta) = \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{j(\beta \sin v - nv)} dv \rightarrow n^{\text{th}} \text{ order, 1st kind Bessel function}$$

• This doesn't depend on  $f_m$ ? wrong these are just the coefficients of Fourier series, finally we have  $\sum_{n=-\infty}^{\infty} J_n(\beta) e^{j2\pi n f_m t}$

$$J_n(\beta) \approx \frac{(\beta/2)^n}{n!} \text{ for } \beta \ll 1$$

• This decays very fast - so let us try to get the 99% bandwidth of the signal from the coefficients

$$99\% \text{ BW} = \sum_{n=-K}^K J_n^2(\beta)$$

$$= J_0^2(\beta) + 2 \sum_{n=1}^K J_n^2(\beta) \rightarrow \text{there is a symmetry}$$

• Bessel functions appear in a lot of places  $\rightarrow$  drum surface vibrations, and a lot of places  $\rightarrow$  find

• ~~Bandwidth  $\approx 2(K+1)$~~

• Carson's Rule  $| K \approx \beta + 1 \rightarrow \text{approx}$

$$\therefore \text{Bandwidth} \approx 2(K+1) \approx 2(\beta+1) f_m \quad \text{Bandwidth of message}$$

• So there will be some distortions, we do not send  $e^{j\beta \sin 2\pi f_m t}$  but  $\sum_{n=-K}^K e^{j2\pi n f_m t}$

• How do you modulate FM signals  $\rightarrow$  Voltage Controlled Oscillator (VCO)

$$\text{say } f(t) = K_v V(t) + f_c \rightarrow \text{we get } \cos(2\pi f_c t + 2\pi K_v \int V(\tau) d\tau)$$

$\downarrow$   
differentiate it

How do you ~~de~~ demodulate the signal, you differentiate it to get the freq outside

we get  $A_c \sin(\dots) \times (2\pi k_f m(t) + 2\pi f_c)$

$\Delta f$  the frequency deviation should not be large to ~~de~~ demodulate

> this is now similar to AM, and we can use that to system

For P-M the analysis is still the same, we get a shift in  $J_n(\beta)$

But ~~the~~ bandwidth changes

$$BW_{FM} \approx 2(\Delta f + f_m) \quad \left| \quad BW_{PM} \approx 2(\Delta \phi + 1)f_m \right. \quad \text{Check}$$

For the same message FM has a ~~less~~ lesser bandwidth

FM or PM is spectrally inefficient compared to AM, but in the power efficiency it is almost 100%. In FM, this becomes significant when noise is introduced (we want high SNR)

New class

$$\beta_{PM} = k_p \max |m(t)|, \quad \beta_{FM} = \frac{k_f \max |m(t)|}{B}$$

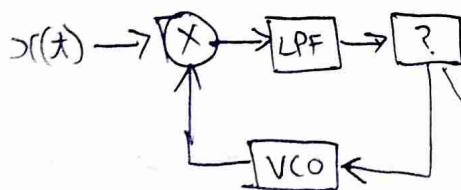
We want a better demodulator than a differentiator for FM

Phase Locked Loop PLL

$$r(t) = A_c \cos(2\pi f_c t + \phi(t)) \rightarrow \text{X} \rightarrow \frac{A_c A_v}{2} \left[ \sin(2\pi f_c t + \phi(t) + \phi_v(t)) + \sin(\phi(t) - \phi_v(t)) \right]$$

VCO  
 $-A_v \sin(2\pi f_c t + \phi_v(t))$

removed from LPF  
 see last page



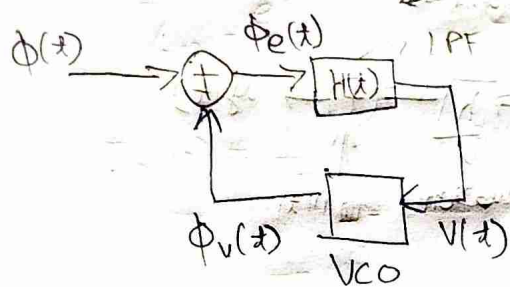
we drive the error to zero

How, we have  $\frac{A_c A_v}{2} (\sin(\phi(t) - \phi_v(t)))$

If this goes to zero then,  $\phi_v(t) \approx \phi(t)$  our ~~de~~ demodulated signal

our message is what is present in  $\phi_v(t)$  → Next page

Assumption  $\approx \frac{A_c A_v}{2\pi} [\phi(t) - \phi_v(t)]$  if error is small



as  $f(t) = f_c + k_v V(t)$  for the VCO

$$\phi_e(t) = \phi(t) - \phi_v(t) \Rightarrow \frac{d\phi_e}{dt} + k_v V(t) = \frac{d\phi(t)}{dt}$$

$$\Rightarrow \frac{d\phi_e(t)}{dt} + 2\pi k_v h(t) \times \phi_e(t) = \frac{d\phi(t)}{dt} \quad \text{Fourier transform}$$

$$j2\pi f \phi_e(f) + 2\pi k_v H(f) \times \phi_e(f) = 2\pi f \bar{\phi}(f)$$

$$\Rightarrow \bar{\phi}_e(f) = \frac{\bar{\phi}(f)}{1 + \frac{k_v H(f)}{j f}}$$

$$\begin{aligned} V(f) &= H(f) \bar{\phi}_e(f) \\ &= \frac{H(f)}{1 + \frac{k_v H(f)}{j f}} \bar{\phi}(f) \end{aligned}$$

assume  $|H(f)| \gg 1$  (put an amplifier)

inside the bandwidth of the

message  $\rightarrow$  so we can use a LPF instead of an amplifier, Gain of LPF should be high

$$\therefore V(f) = \frac{j f}{k_v} \bar{\phi}(f)$$

$$V(t) = \frac{1}{2\pi k_v} \frac{d\phi(t)}{dt} = \frac{1}{k_v} m(t) \Rightarrow V(t) \text{ is proportional to our message}$$

$\hookrightarrow$  this is a PI controller, VCO is the integrator block

• Pg 128 of text book has a lot more content on PLL

new class

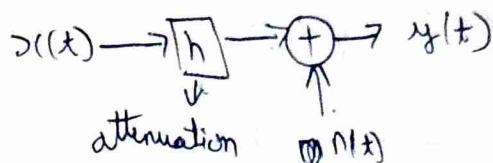
• PLL is unstable  $\rightarrow$  the message should be bandlimited

• If our message is a DC component, the integrator causes the system to become unstable

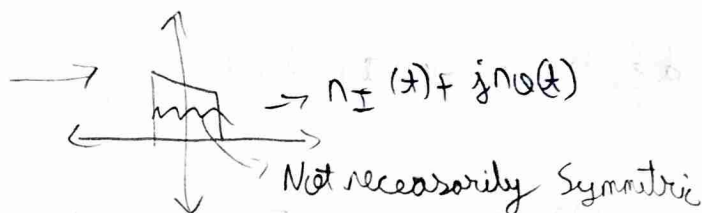
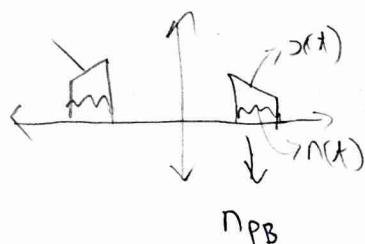
1) Would you pick AM or FM?

2) Let us look at Additive Noise in channels

$$y = hx + n$$



3) There is noise in baseband and passband, but the passband noise is real, but baseband noise ~~is~~ is not necessarily real



4) Our LPF determines the quality of noise effect.

5) In AM, if

|                           | AM     | FM             | AM              | FM                      |
|---------------------------|--------|----------------|-----------------|-------------------------|
| Noise                     |        |                |                 |                         |
| $M_{\text{demod}} I(t) =$ | $m(t)$ | $\cos \phi(t)$ | $m(t) + n_I(t)$ | $\cos \phi(t) + n_I(t)$ |
| $M_{\text{demod}} Q(t) =$ | 0      | $\sin \phi(t)$ | $n_Q(t)$        | $\sin \phi(t) + n_Q(t)$ |

6) Demodulation is simpler for FM  $\rightarrow$  take derivative

at the receiver let's say we have

$$\cos \phi(t) + n_I(t) + j(\sin \phi(t) + n_Q(t))$$

↓ combine terms

we get  $e^{j\phi(t)} + n(t) e^{j\theta(t)} \rightarrow$  we project this signal into the unit circle in the ~~real~~ complex plane,

Our message  $e^{j\phi(t)}$  will always be found there while noise can be anywhere.

7) We do this as our message signal is contained in the phase.

## New class

FM has a better power efficiency  $\rightarrow$  which makes it ~~more~~ robust to noise

$$y(x) = \underbrace{h}_{\substack{R_x \\ \downarrow \\ \text{attenuation}}} \underbrace{x(x)}_{T_x} + \underbrace{n_{RF}(x)}_{\substack{\downarrow \\ \text{noise}}} \rightarrow n_I(x) + j n_Q(x)$$

## Random Variables

$X(t) \rightarrow$  random variable

$\hookrightarrow$  Sample Space,  $\mathcal{F} \rightarrow$  event space, Probability Measure

$$\mathcal{X} = \{H, T\}$$

$$2^{\mathcal{X}}$$

$$P \rightarrow \text{P.D.F}$$

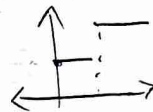
$$X: \mathcal{X} \rightarrow \mathbb{R}^n$$

Borel Set  $\rightarrow$  Our message is a random variable, but  $\mathcal{X}$  is uncountably  $\infty$  so you use this to pick model probability

PDF for a few random variable may not even exist, but CDF exists

$\hookrightarrow$  HW

CDF  $\rightarrow$  monotonic, Right Continuous



$$F_X(x) = P(X \leq x)$$

$$E[g(X)] = \int g(x) f_X(x) dx$$

$$\text{Var}(X) = E\{(X - E[X])^2\}$$

$$\text{Cov}(X, Y) = E\{(X - E[X])(Y - E[Y])\}$$

~~Cov~~ Cross Covariance  
Cov  $\rightarrow$  real

If  $\otimes$  the Random Variable is complex you will do conjugate

$$\text{Var}(X) = E[(X - E[X])^* (X - E[X])] \rightarrow \text{this is real}$$

$$\text{Cov}(X, Y) = E[(X - E[X])^* (Y - E[Y])] \rightarrow \text{this can be complex}$$

$$\text{Bernoulli} \rightarrow X \in \{x_0, x_1\} \xrightarrow{P} \text{Be}(P) \xrightarrow{1-P}$$

$$\text{Uniform} \rightarrow X \sim U[a, b], f_X(x) = \begin{cases} \frac{1}{b-a} \\ 0 \end{cases}$$

1) Gaussian  $\rightarrow X \sim \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$

2)  $f_{X,Y}(x,y) = f_X(x) f_Y(y) \rightarrow$  means Independent

3)  $E[X,Y] = E[X]E[Y] \rightarrow$  Uncorrelated

4) Gaussian R.V are ~~uncorrelated~~ Uncorrelated and Independent

$Z = \begin{bmatrix} X \\ Y \end{bmatrix} \rightarrow \frac{1}{2\pi \det(\Sigma)} e^{-\frac{1}{2}(Z-\mu)^H \Sigma^{-1}(Z-\mu)}$

If  $X_{n \times 1}$   $\rightarrow \Sigma^{n \times n}$

$\Sigma = \begin{bmatrix} \sigma_{11} & \dots & \sigma_{1n} \\ \vdots & \ddots & \vdots \\ \sigma_{n1} & \dots & \sigma_{nn} \end{bmatrix}$   
 $\sigma_{ij} = \text{cov}(X_i, X_j)$

$\begin{bmatrix} N_I(t) \\ N_Q(t) \end{bmatrix} = N \begin{matrix} \nearrow N(0, \sigma_I^2) \\ \searrow N(0, \sigma_Q^2) \end{matrix}$

$\rightarrow \frac{1}{2\pi \det(\Sigma)} e^{-\frac{1}{2} \left( \begin{bmatrix} N_I \\ N_Q \end{bmatrix}^T \Sigma^{-1} \begin{bmatrix} N_I \\ N_Q \end{bmatrix} \right)}$   
 $\det(\Sigma) = \sigma_I^2 \sigma_Q^2$   
 $\Sigma^{-1} = \begin{bmatrix} \frac{1}{\sigma_I^2} & 0 \\ 0 & \frac{1}{\sigma_Q^2} \end{bmatrix}$   
 $\frac{N_I^2}{\sigma_I^2} + \frac{N_Q^2}{\sigma_Q^2}$

↓  
They are independent  
 $\Sigma = \begin{bmatrix} \sigma_I^2 & 0 \\ 0 & \sigma_Q^2 \end{bmatrix}$

5) If you transform  $N$  to polar coordinates what happens to the distribution of  $\begin{bmatrix} r \\ \theta \end{bmatrix}$  from  $\begin{bmatrix} N_I \\ N_Q \end{bmatrix}$

$R = \sqrt{N_I^2 + N_Q^2}$   $R = \sqrt{N_I^2 + N_Q^2}$ ,  $\Theta = \tan^{-1}\left(\frac{N_Q}{N_I}\right)$

6)  $Y = g(X)$ ,  $f_Y(y) = \frac{f_X(g^{-1}(y))}{|g'(y)|} = \frac{f_X(g^{-1}(y))}{\det(J)}$   
 $\rightarrow$  for multivariable

$N_I = R \cos \Theta$

$N_Q = R \sin \Theta$ , find Jacobian of this

HW, find  $f_{R,\theta}(r,\theta) = ?$

New class

Random Vector

If  $X \in \mathbb{R}^n$ ,  $X \sim N(\mu, \Sigma)$

$\nearrow$  Co-variance matrix

$$= \frac{1}{(2\pi)^{n/2} \det(\Sigma)} e^{-\frac{1}{2} (x-\mu)^T \Sigma^{-1} (x-\mu)}$$

$$\text{If } X \in \mathbb{C}^n = \frac{1}{(2\pi)^n \det(\Sigma)} e^{-\frac{1}{2} (x-\mu)^H \Sigma^{-1} (x-\mu)}$$

If  $\Sigma$  is diagonal matrix  $\rightarrow$  ~~we~~ distribution is independent  
so multiply the marginals to get joint P.O.F

Yesterday we were doing  $X \rightarrow Y$   
 $\begin{bmatrix} n_I \\ n_Q \end{bmatrix} \rightarrow \begin{bmatrix} r \\ \theta \end{bmatrix}$

$\nearrow$  Pg 218 in textbook

$$J = \begin{bmatrix} \frac{\partial r}{\partial n_I} & \frac{\partial r}{\partial n_Q} \\ \frac{\partial \theta}{\partial n_I} & \frac{\partial \theta}{\partial n_Q} \end{bmatrix} = \begin{bmatrix} \frac{n_I}{\sqrt{n_I^2 + n_Q^2}} & \frac{n_Q}{\sqrt{n_I^2 + n_Q^2}} \\ -\frac{n_Q}{\sqrt{n_I^2 + n_Q^2}} & \frac{n_I}{\sqrt{n_I^2 + n_Q^2}} \end{bmatrix}$$

you have to divide by  $\det(J)$  for your distribution  
 $\det(J) = \frac{1}{r}$   $\rightarrow$  If you are doing a transformation

$$f_{r,\theta}(r,\theta) = \frac{r}{2\pi\sigma^2} e^{-\frac{1}{2} \left(\frac{r^2}{\sigma^2}\right)}$$

$$f_r(r) = \int_{-\pi}^{\pi} (\dots) d\theta = \frac{r}{\sigma^2} e^{-\frac{1}{2} \left(\frac{r^2}{\sigma^2}\right)} \rightarrow \text{Rayleigh distribution}$$

$$f_\theta = \frac{f_{r,\theta}}{f_r} = \frac{1}{2\pi} \rightarrow \text{they are uniform, } \theta \sim \text{Unif}[-\pi, \pi]$$

$$R \sim \frac{r}{\sigma^2} e^{-\left(\frac{r^2}{\sigma^2}\right)}$$

$\nearrow$  distribution in this direction

$$\begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} n_I \\ n_Q \end{bmatrix} \rightarrow \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{n_I^2}{2\sigma^2}}$$

What about a distribution in any other direction  $\rightarrow$  we get the same distribution  $\rightarrow$  Symmetric around 0

1) They are called Circular Symmetric Complex Gaussian

2) If  $\sigma_0 \neq \sigma_I$  it is not symmetric

3)  $y(t) = h x(t) + n(t)$

but our  $n(t)$  is a random process  $X_1, X_2, \dots, X_n \rightarrow$  a sequence  
but we have continuous time random process,  $\therefore X(t)$

4) For any random variable we look at its statistics (Mean etc)

5) For a random process  $\rightarrow X_{t_1} - X_{t_2}$

different time same statistic  $\rightarrow$  strict sense Stationary process

In real life Wide sense stationary - (WSS)

1)  $\downarrow$  mean is stationary  $m_X(t) = E\{X(t)\} = m_X$  (constant)

$\downarrow$   
This is a signal  
deterministic

Auto correlation

2)  $R_X(t_1, t_2) \rightarrow$  correlation of random process  
 $\hookrightarrow$  not auto correlation

$\uparrow$   
 $= E[X(t_1) X^*(t_2)]$   
 $= \int X(t_1) X^*(t_2) f(X(t_1), X(t_2)) dt_1 dt_2$

dependence on  $t_1 - t_2$  not  $t_1$  or  $t_2$

$\uparrow$   
if this  $= R_X(t_1 - t_2) \rightarrow$  (WSS)

$\hookrightarrow$  Samples separated by same amount of time have a similar distribution

3)  $X(t)$   $\rightarrow m(t)$   
WSS  $\rightarrow R(\tau)$

we can see  $R(\tau) = R^*(\tau)$

$\hookrightarrow$  will have real spectrum

4) Example of random process  $X(t) = A_c \cos(2\pi f_c t + \theta)$

$\downarrow$   
Unif  $\sim [-\pi, \pi]$

$$m_X(t) = E[X(t)] = \frac{1}{2\pi} \int_{-\pi}^{\pi} A_c \cos(2\pi f_c t + \theta) d\theta = 0 \rightarrow \text{it is stationary}$$

$$\circ) R_x(t_1, t_2) = \int \cancel{x(t_1)} \cancel{x^*(t_2)} \mathbb{E} \{ \cancel{x(t_1)} \cancel{x^*(t_2)} \}$$

$$= \int A_c^2 \cos(2\pi f_c t_1 + \theta) A_c \cos(2\pi f_c t_2 + \theta) dx_{t_1} dx_{t_2}$$

here  $\theta$  is random but it is fixed (doesn't change with time)  
 $\rightarrow$  this is WSS  $\rightarrow$  How?

New class

$\hookrightarrow$  for analog systems we have continuous time random process

$$\begin{array}{ccccc} Y(t) & = & hX(t) & + & N(t) \\ \downarrow & & \downarrow & & \downarrow \\ R_x & & T_x & & \text{AWGN} \end{array}$$

$$\circ) \text{WSS} \doteq m_X(t) = m_X \quad \left| \quad R_{XX} = R_{XX}(t_1 - t_2) \right. \quad \nearrow \text{lag or delay}$$

$\circ$  Example from last class

$$X(t) = A_c \cos(2\pi f_c t + \theta) \quad \hookrightarrow \text{Unif}[-\pi, \pi]$$

$= g(\theta, t) \rightarrow \theta$  is fixed after being picked randomly  
 but different  $t$  gives a new value  $\rightarrow$  this  
 is a random process

$$\circ) \mathbb{E} \{ x(t) \} = ?$$

$$\begin{aligned} \hookrightarrow \bullet \mathbb{E} \{ g(\theta) \} &= \int g(\theta) f_\theta(\tau) d\tau \quad \phi = g(\theta) \\ &= \int \phi f_\phi d\tau \quad \rightarrow \text{Prove this} \end{aligned}$$

$$f_\phi = \frac{f_\theta(g^{-1}(\phi))}{g'} \quad f_\theta(\theta)?$$

from last page

$$\mathbb{E}\{X(t)\} = \int_{-\pi}^{\pi} A_c \cos(2\pi f_c t + \theta) \frac{d\theta}{2\pi} = 0$$

$$\mathbb{E}\{R_{XX}(t)\} = \iint A_c \cos(2\pi f_c t_1 + \theta) A_c \cos(2\pi f_c t_2 + \theta) f_{\theta}(x_{t_1}, x_{t_2}) dx_{t_1} dx_{t_2}$$

1) We are taking ensemble average  $\uparrow$ , the process is ergodic

→ try transforming to  $g(\theta)h(\theta)$  and integrate over  $\theta$

2) We transform and integrate over  $\theta$

we get  $R_{XX}(t) = \frac{A_c^2}{2} \cos(2\pi f_c (t_1 - t_2))$

3) Say during transmission  $X(t) = M(t) \cos 2\pi f_c t$

↓  
Say it is  
WSS

$$\mathbb{E}\{X(t)\} = \mathbb{E}\{M(t)\} \cos 2\pi f_c t \rightarrow \text{periodic}$$

4) But at the receiver  $X(t) = M(t) \rightarrow$  which is again WSS

→ We see blocks in comsys change the stationary state

what is  $R_{XX}$  of this?

$$R_{XX}(t) = \mathbb{E} \left\{ M(t_1) \cos 2\pi f_c t_1 M^*(t_2) \cos 2\pi f_c t_2 \right\}$$

$$= \mathbb{E} \{ M(t_1) M^*(t_2) \} \{ \cos(2\pi f_c (t_1 - t_2)) + \cos(2\pi f_c (t_1 + t_2)) \}$$

mean and  $R_{XX}$   
is periodic

④ Cyclostationary → the above this is periodic

→ How do you get this in a computer  
How to simulate

5) Sensors use this (CSMA) to see if a band is occupied or not

•  $X(t)$  is periodic with  $\frac{1}{k_c}$

• How to say it is periodic  $R_{XX}(t_1 + T_1, t_2) = R_{XX}(t_1, t_2)$   
 $= R_{XX}(t_1, t_2 + T_2)$

$T_1 = T_2$  for Cyclostationary

• What happens to the statistic of a random process after going through an LTI system

$$X(t) \xrightarrow[h(t)]{\boxed{\text{LTI}}} Y(t) \quad Y(t) = h(t) \otimes X(t)$$

$$\mathbb{E}\{Y(t)\} = \mathbb{E}\left\{\int X(\tau) h(t-\tau) d\tau\right\} \quad \leftarrow \text{looking at mean first}$$

• we can swap integrals if integrals are defined expectation

$$\Rightarrow \int \mathbb{E}(X(\tau)) h(t-\tau) d\tau$$

$$\mathbb{E}\{Y(t)\} = m_X(t) \int_{-\infty}^{\infty} h(\tau) d\tau$$

$\downarrow$  Not random       $t-\tau$  doesn't matter  
 $\rightarrow$  not same as  $\int_{-\infty}^{\infty} |h(\tau)| d\tau < \infty$

if  $\otimes$  system is "stable". our expectation becomes  $\infty$ .  
 not

• ~~Defining~~ defining cross correlation

$$R_{XY}(t_1, t_2) = \mathbb{E}\{X(t_1) Y^*(t_2)\}$$

• if  $X, Y$  are WSS and if  $R_{XY}(t_1, t_2) \xrightarrow{\tau}$   
 they are jointly WSS  $\uparrow$

• We again have a property  ~~$R_{XY}(\tau) = R_{YX}^*(-\tau)$~~   $R_{XY}(\tau) = R_{YX}^*(\tau)$

$\hookrightarrow$   $X$  and  $Y$  are even and real then  $R_{XY} = R_{YX}$

New class

$$R_{YX} = E \left\{ X(t+\tau) Y(t) \right\} \rightarrow \text{if } X, Y \text{ are real}$$

$$R_{YX} = E \left\{ Y(t+\tau) X(t) \right\}$$

~~EEE~~

1) when are the above things equal  $\rightarrow$  When again?

2) how do you say two random processes variables are same

1  $\hookrightarrow$  Strict equality  $\rightarrow$  basically duplicate

2  $\hookrightarrow$  Almost sure  $X(t) = Y(t)$  when  $P(t) \neq 0$

3  $\hookrightarrow$  Same in distribution  $X$  and  $Y$  can represent multiple things but their distribution is same.

$\rightarrow$  we are taking about 2,3 in our case of  $R_{XY}$

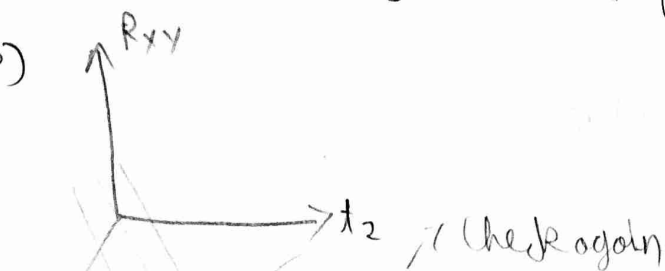
$$Y(t+\tau) X(t) \rightarrow Y(-t-\tau) X(-t)$$

$$\downarrow \quad \downarrow \quad \Rightarrow \quad Y(v) X(v+\tau)$$

$\downarrow$   
equal to above first thing

1)  $X$  was  $\tau$  ahead of  $Y$ ,  $\downarrow$  Some here after flipping

2)



$\tau$  line, we only see a 1-D curve instead of 3-D

$\downarrow$  it is some along here (blue lines)

$$\circ) R_{xx}(\tau) = \mathbb{E} \{ X(t+\tau) X(t) \} \rightarrow \text{for real}$$

$$R_{xx}(-\tau) = \mathbb{E} \{ X(t-\tau) X(t) \}$$

$$\circ) X(t) \rightarrow \boxed{\text{LTI}} \xrightarrow{h(t)} Y(t)$$

$$R_{yy}(t_1, t_2) = \mathbb{E} \left\{ X^*(t_2) \underbrace{X(t_1-\tau) h(\tau)}_{Y(t_1)} d\tau \right\}$$

$$= \int \mathbb{E} \{ X(t_1-\tau) X^*(t_2) \} h(\tau) d\tau$$

$$R_{yy}(\tau) = R_{xx}(\tau) * h(\tau) \quad R_{yx} = R_{xx}(v) * h(v)$$

$$= R_{yx}^*(-\tau)$$

$$\circ) R_{xy} = R_{xx}^*(-v) \oplus h(-v) \rightarrow \text{statistics of random process are also convolved with } h(\tau) \rightarrow \text{Because of linearity}$$

$$= R_{xx}(v) \oplus h(-v)$$

$$\circ) R_{yy}(\tau) = R_{xx} * h(v) * h^*(-v)$$

New class  $\rightarrow$  are WSS

$$\circ) X(t), Y(t) \text{ and } R_{xy}(t_1, t_2) = R_{xy}(\tau) = R_{yx}^*(-\tau) \text{ then}$$

$X, Y$  are jointly WSS

$$\circ) R_{yx} = R_{xx}(\tau) * h(\tau) \rightarrow \text{this doesn't tell us } Y(t) \text{ is WSS}$$

$$\circ) R_{xy} = R_{xx}(\tau) \oplus h^*(-\tau)$$

$$\circ) R_{yy} = \{ Y(t_1) Y^*(t_2) \}$$

$$= \mathbb{E} \int X(t_1-\tau) h(\tau) d\tau Y^*(t_2)$$

$$= \int R_{xy}(t_1-\tau-t_2) h(\tau) d\tau$$

$$= \int R_{xy}(v-\tau) h(\tau) d\tau$$

$$R_{yy} = R_{xy}(v) \oplus h(v) = R_{xx}(v) * h^*(-v) * h(v)$$

$$\cdot) R_{yy} = R_{xx}(v) * h^*(v) * h(v)$$

•) Looking at the spectrum of a random process

•) Say  $X \in \mathbb{R}^n \sim N(\mu, \Sigma)$  gaussian ~~with~~ random vector  
 $\downarrow$   
 Co-variance

•)  $Y = F X$   
 $\downarrow$   
 DFT Matrix,  $Y$  is also gaussian  $\sim N(F\mu, F\Sigma F^H)$

If  $\Sigma = \sigma^2 I \rightarrow$  i.i.d

then the distribution of  $Y$  is also gaussian

$F$  is a unitary matrix (Just rotates)

•) Instantaneous spectrum is just random.

•)  $E\{|X(f)|^2\} \rightarrow$  PSD (Power Spectral Density)

$= E\{R_{xx}(\tau)\} \rightarrow$  Wiener-Khinchin Theorem  
 $\hookrightarrow$  if it is WSS

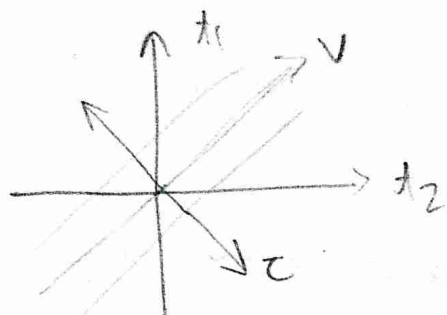
•) ~~Oscilloscope~~ Oscilloscopes take and show PSD  $\rightarrow$  they cannot show phase of Fourier transform.

•) When does the Fourier transform guarantee real and positive output? as we are doing  $E\{|X(f)|^2\}$   
 $\downarrow$   
 Conjugate Symmetric

$$\begin{aligned} |X(f)|^2 &= X(f) X^*(f) \\ &= \int X(t_1) e^{-j2\pi f t_1} dt_1 \int X^*(t_2) e^{+j2\pi f t_2} \\ &= \int X(t_1) X^*(t_2) e^{-j2\pi f (t_1 - t_2)} \end{aligned}$$

$$E\{|X(f)|^2\} = \iint R_{xx}(\tau) e^{-j2\pi f \tau} dt_1 dt_2$$

•)  $x(t)$   $x^*(-t)$   $\rightarrow x(t) \times x^*(-t)$  will always have a non negative transform



transform from  $t_1, t_2$  to  $\tau, v$   
 $\tau \perp v$

we get

$$\begin{bmatrix} \tau \\ v \end{bmatrix} = \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} t_1 \\ t_2 \end{bmatrix}$$

$$= \iint R_{xx}(\tau) e^{j2\pi f \tau} \frac{d\tau \cdot dv}{2} \rightarrow \text{what about limits?}$$

$t_1 - t_2$ ,  $\lim_{t_1, t_2 \rightarrow \infty}$  is what?

•) But here we get  $\infty$  in integration

•) So we normalize

$$\lim_{T \rightarrow \infty} \frac{1}{T} \mathbb{E} \{ |x_T(f)|^2 \}, \text{ where } x_T(f) = \int \{ x(t) \cdot \text{Rect}(\frac{t}{T}) \}$$

$\downarrow$   
This is our definition of PSD



$$•) N(t) \sim N(0, \sigma^2)$$

$$m_N(t) = 0, R_{NN}(t_1, t_2) = \delta(t) \quad t_1 \neq t_2 \quad \mathbb{E} = 0$$

Qing 1

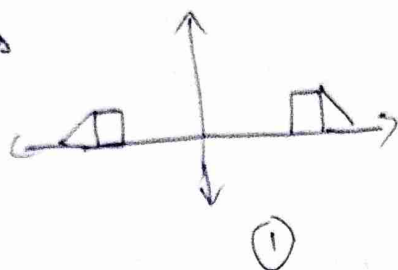
$$Q1 \quad \frac{1}{2\pi} \frac{d\phi(t)}{dt} = f(t) = \phi(t)$$

Q2  $\rightarrow$  can have a ~~highpass~~ highpass filter at a higher freq

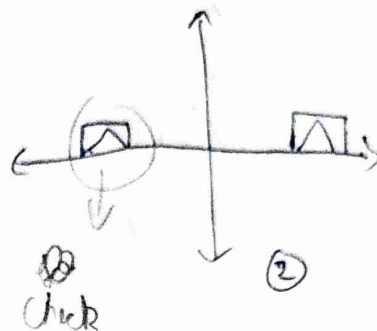
$\rightarrow$  Consider carrier freq as zero  $\rightarrow$  modulate f FM signal with that assumption. hiltbert transform

$$x(t) = (0 + k_f \int_0^t m(\tau) d\tau) \rightarrow \text{then transport } x(t) + j \hat{x}(t)$$

### Q3 Signals



vs



Both have ~~low~~ some bandwidth efficiency.

In real life we use (2), because hardware is simple. Only one oscillator

Q5 In signal power  $\rightarrow$  if signals are orthogonal you can just add their ~~power~~ power

Q6  $Y(t) = M(t) \cos(2\pi f_c t + \theta) + N(t)$

$\downarrow$

WSS

$\downarrow$

we know it is WSS

$\downarrow$

Unif

$\downarrow$

i.i.d  $N(0, \sigma^2)$

$\underbrace{\hspace{10em}}$

$R_{XX}(\tau)$

$\underbrace{\hspace{10em}}$

also WSS

$\therefore Y(t) \sim \mathbb{R}$  they are independent so you can add them

$\therefore Y(t)$  is WSS

1) Ideal ~~LPF~~ LPF,  $h(t) * Y(t) \cos 2\pi f_c t$  is still WSS

### 2) Newclass

Why is Autocorrelation a Power signal  $\rightarrow Z_t$  is a 2D signal only for a WSS signal. ~~which~~ it is a 1D signal

1) Auto correlation ~~is~~ function can never be zero  $\rightarrow$  implies energy ~~is~~ is zero.

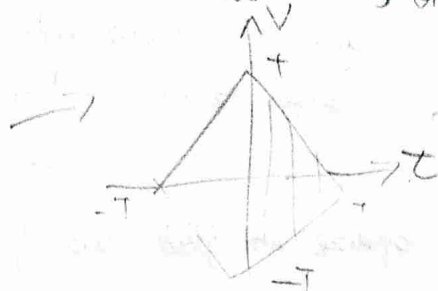
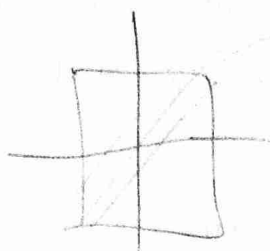
1) Is there a signal where  $\int R_{XX}(\tau) d\tau = 0 \rightarrow \cos(2\pi f_c \tau) \rightarrow$  in the limit of  $\infty$

2) We look at autocorrelation in a ~~random~~ window, for defining PSD

we have  $\lim_{T \rightarrow \infty} \frac{1}{T} E\{|X_T(t)|^2\} = \int_{-T/2}^{T/2} \int_{-T/2}^{T/2} R_{XX}(t_1 - t_2) e^{-j2\pi f_c(t_1 - t_2)} dt_1 dt_2$

$$= \lim_{T \rightarrow \infty} \frac{1}{T} \iint_{-T}^{T+|z|} R_{xx}(\tau) e^{-j2\pi f \tau} \frac{d\tau dz}{2}$$

Not WSS, only when  $T \rightarrow \infty$



→ We have just rotated it

$$z = t_1 - t_2$$

$$v = t_1 + t_2$$

New class

Wiener-Kin Khinchin Theorem

PSD of  $X(t) = \mathcal{F}\{R_{xx}(\tau)\}$

will it work if  $X$  is SSS → Yes (Because it is still WSS)

SSS says their  $(X_1(t_1), X_2(t_2))$  have the same distribution  $X_1(t_1-d)$

we had then  $\lim_{T \rightarrow \infty} E \left\{ \frac{1}{T} \int_{-T/2}^{T/2} \int_{-T/2}^{T/2} X_1(t_1) e^{-j2\pi f t_1} X_2(t_2) e^{j2\pi f t_2} dt_1 dt_2 \right\}$

limits come from  $\text{rect}(t/T)$

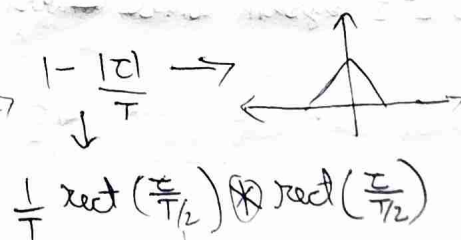
but  $X_1(t) = X(t) \text{rect}(t/T)$  → this is not WSS

1)  $X_1(t)$  and  $R_{xx}$  need to satisfy Dirichlet conditions → then we can take the expectation inside.

$$\Rightarrow \lim_{T \rightarrow \infty} \frac{1}{T} \iint_{-T}^{T+|z|} R_{xx}(\tau) e^{-j2\pi f \tau} \frac{d\tau dz}{2}$$

$$= \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T}^T R_{xx}(\tau) e^{-j2\pi f \tau} d\tau \times \left( \frac{2T - 2|z|}{2T} \right)$$

Fourier transform



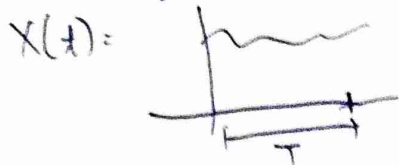
$$\frac{1}{T} \text{rect}\left(\frac{\tau}{T}\right) \otimes \text{rect}\left(\frac{z}{T}\right)$$

check

$$\lim_{T \rightarrow \infty} R(f) \times T \text{sinc}^2(fT)$$

as  $T \rightarrow \infty$ , this tends to  $\delta(f)$ , ∴ we get  $R(f) \times \delta(f) \rightarrow R(f)$

How to find PSD & IRL



take  $\int X(t) X^*(t-\tau)$  in different windows  
and correlate with it self with the  
some delay  $\rightarrow$  then average over  
This is the expectation

Small window may not capture the ~~full~~ low freq signals in that window, large windows are a ~~revers~~ ~~a~~ ~~problem~~  $\rightarrow$  you do not need to go beyond the lowest freq.

AWGN Model

$$Y(t) = X(t) + N(t), \quad N(t) \sim N(0, \sigma^2)$$

$$\therefore R_{NN}(\tau) = \sigma^2 \delta(\tau) \rightarrow \tau \rightarrow$$

it is present in all frequencies  $\rightarrow$  white

But this has  $\infty$  power  $\rightarrow$  SNR = 0

$$SNR = \frac{\text{Power } X(t)}{\text{Power } N(t)}$$

In real life there is always a filter in our hardware so we never observe the full spectrum of the Noise

$$Y(t) = (X(t) + N(t)) * \vec{h(t)} \text{ filter}$$

We are starting with an ~~uncorrelated~~ ~~uncorrelated~~ uncorrelated function  
 $\rightarrow \delta(\tau)$  but after you filter the correlation changes.

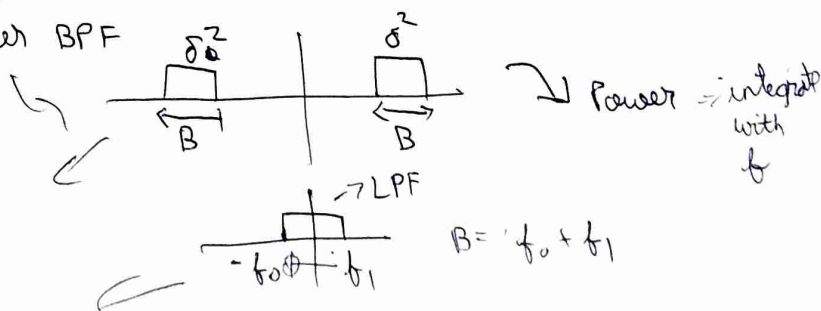
The samples are still from gaussian distribution but they are correlated

LTI System Cannot Convert an uncorrelated signal to correlated

Spectrum of noise after BPF

$$\text{Power} = \sigma^2 B$$

$$\text{Power} = \sigma B$$



New class

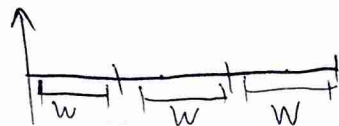
Why do we define  $R_{NN}(\tau) = \delta(\tau)$  instead of  $\begin{cases} 1 & \tau=0 \\ 0 & \tau \neq 0 \end{cases}$

↓  
not  
Power is zero

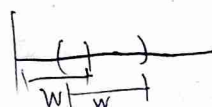
↓  
you get Power 0  
lim = 0 at  $\tau=0$   
But we have 1 at  $\tau=0$

We are looking at Bandlimited Power spectrum so we want it to be  $\delta(\tau)$

Computing PSP



→ Take FFT for W samples and do average (Bartlett)



→ overlapped window (Welch)  
then take FFT

IRL



$$R_{xx}(m) = \frac{1}{N} \sum_{i=1}^N x[i] x[i-m] \rightarrow \text{after this do fft.}$$

Say  $R_{NN}(\tau) = N_0 \delta(\tau)$ ,  $N_0 = kT \rightarrow \text{Noise}$

Noise Power =  $N_0 B$  → Bandwidth (including -ve freq)

An amplifier reduces SNR



$$\sqrt{G} y(t) = \sqrt{G} x(t) + \underbrace{\sqrt{G} N_1(t)}_{P_S}$$

$$y(t) = \sqrt{G} x(t) + N_1(t)$$

→ Noise added by amplifier

$$SNR_{\text{output}} = \frac{G P_S}{G N_0 B + N_1 B}$$

→ power added → Independent.

$$R_{yy} = E \left\{ (x(t_1) + N(t_1)) (x^*(t_2) + N^*(t_2)) \right\}$$

FFT ↓  
 $R_{yy} = R_{xx} + R_{NN} + R_{xN} + R_{Nx}$

$$S_y = S_x + S_N + C_{xN} + C_{Nx} \rightarrow \text{Cross power spectral density}$$

Can be complex as well ( $R_{xy}(\tau) = R_{yx}^*(-\tau)$ )

$C_{xN} + C_{Nx}$  together in real

Check

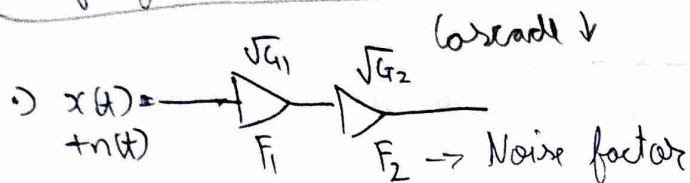
→ signal & noise

• Noise figure Noise factor =  $\frac{SNR_{i/p}}{SNR_{o/p}} \geq 1$

• Noise figure =  $F = \frac{SNR_{i/p}}{10 \log_{10}(P)}$   
(dB)

write dBm  
dBW

• Open WRT to change transmit power of any router



$\sqrt{G_1} x(t) + \sqrt{G_1} n_1(t)$   
 $\sqrt{G_1 G_2} x(t) + \sqrt{G_1 G_2} n_1(t) + \sqrt{G_2} n_2(t)$

•  $SNR = \frac{G_1 G_2 P_s}{G_1 G_2 N_0 B + G_2 N_1 B + N_2 B}$   
output

$F = \frac{SNR_{i/p}}{SNR_{o/p}} = \frac{P_s / N_0 B}{\frac{G_1 G_2 P_s}{G_1 G_2 N_0 B + G_2 N_1 B + N_2 B}} = \frac{1}{N_0 G_1 G_2 B} (G_1 G_2 N_0 B + G_2 N_1 B + N_2 B)$   
 $= 1 + \frac{N_1}{N_0 G_1} + \frac{N_2}{N_0 G_1 G_2}$

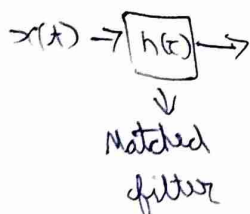
• for a single device,  $F_1 = \frac{P_s / N_0 B}{\frac{G_1 P_s}{G_1 N_0 B + N_1 B}} = 1 + \frac{N_1}{G_1 N_0}$

$F_2 = 1 + \frac{N_2}{G_2 N_0} \rightarrow$  Considering  $N_0$  is given in input

using this we get  $F_1 + \frac{F_2 - 1}{G_1}$

$F_{\text{cascaded}} = F_1 + \sum_{i=2}^n \frac{F_i - 1}{\prod_{j=1}^{i-1} G_j}$  Friis formula

• Can we use some LTI block to maximize SNR



$SNR = \frac{P_x}{P_n} = \frac{R_{xx}(0)}{N_0 B}$   
 $\int_{-\infty}^{\infty} S_x(f) df = R_{xx}(0)$